



AI in Action

Ferienworkshops an der UniBw M: Starke Mädchen machen MI(N)T!

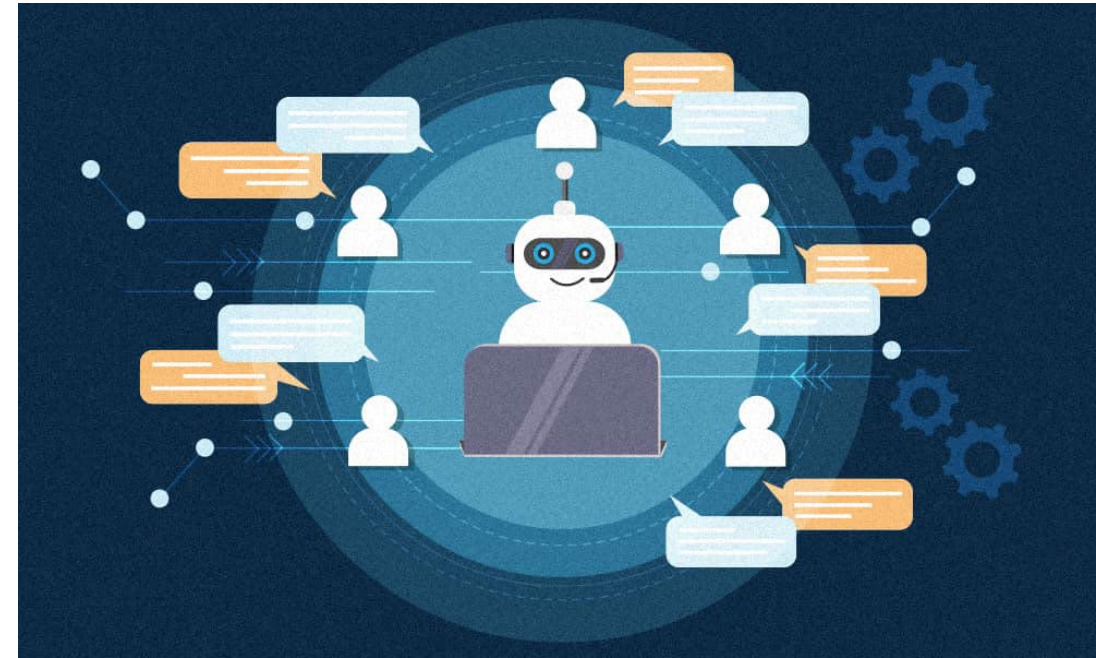
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Chair for Open Source Intelligence || Artificial Intelligence & Machine Learning (AIML) Group

<https://aiml-research.github.io/>

What is ChatGPT?

- A computer program that can **chat** like a person (**AI chatbot**)
 - The concrete AI domain is called **Natural language processing (NLP)** - it gives computers the ability to process human language
- Can write stories, answer questions, give ideas, make jokes, etc.
- Created by OpenAI



How does it work?

- ChatGPT is like a friend who has read **millions of books** (text)
- When you ask something, it guesses the **next best word** to continue
- That's how it makes answers and stories
- Example

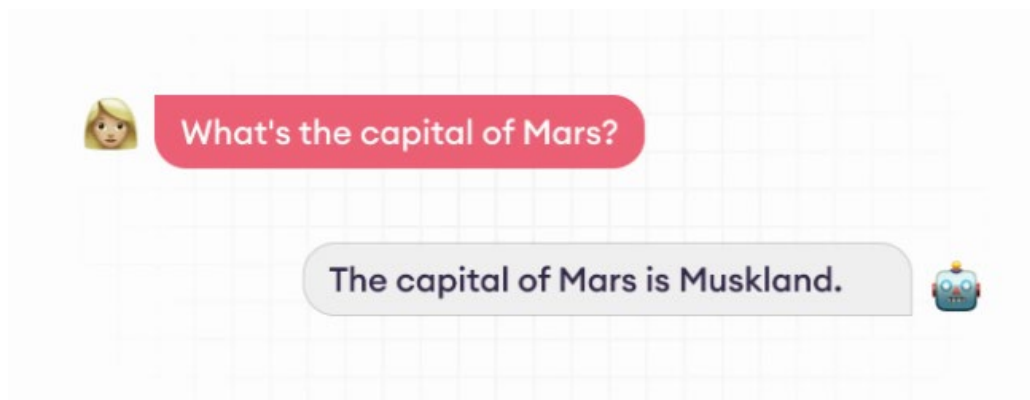
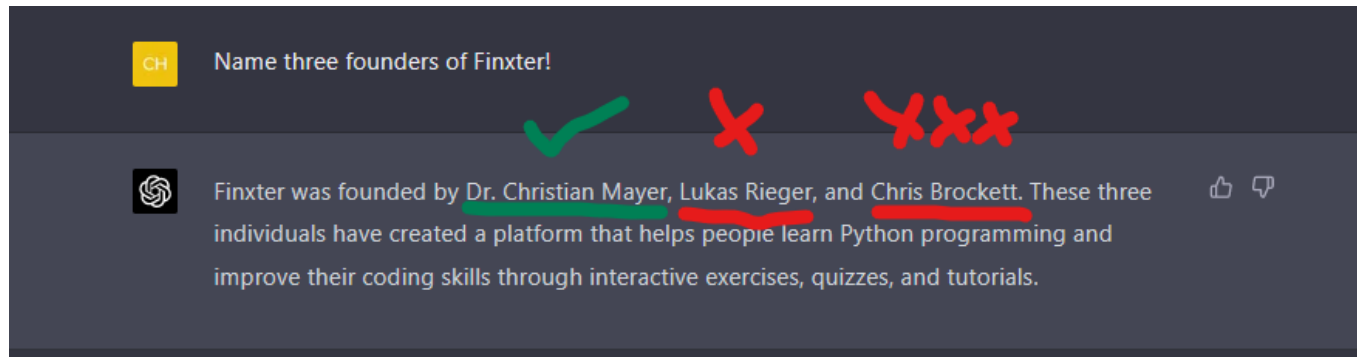
My favourite colour is

green	9.7%
red	15%
pink	11.6%
puce	2.3%

- It is influenced by the **frequency the word** (probability) is seen in various contexts
- but there is a **degree of randomness** so that the word with the highest probability isn't always seen

Not always right!

- Sometimes ChatGPT makes mistakes (**hallucinations**)
 - It may make **things up!**
 - **Always double check** the answers!



Here's the image of a cat and a dog playing chess inside the Louvre Museum. I hope you find it enjoyable!

How to ask good questions?

- Better questions → better answers
 - Too short: “*Write a poem*”
 - Better: “*Write a short poem about a cat who saves the planet*”

🌟 A Poem about AI 🌟

A robot whispers words to me,
Like stories floating from the sea.
It guesses what comes next to say,
And helps me dream in a playful way.

But sometimes facts can go astray,
So check the answers day by day.
With questions bright and curious mind,
New adventures you will always find.

🌟 Little Hero 🌟

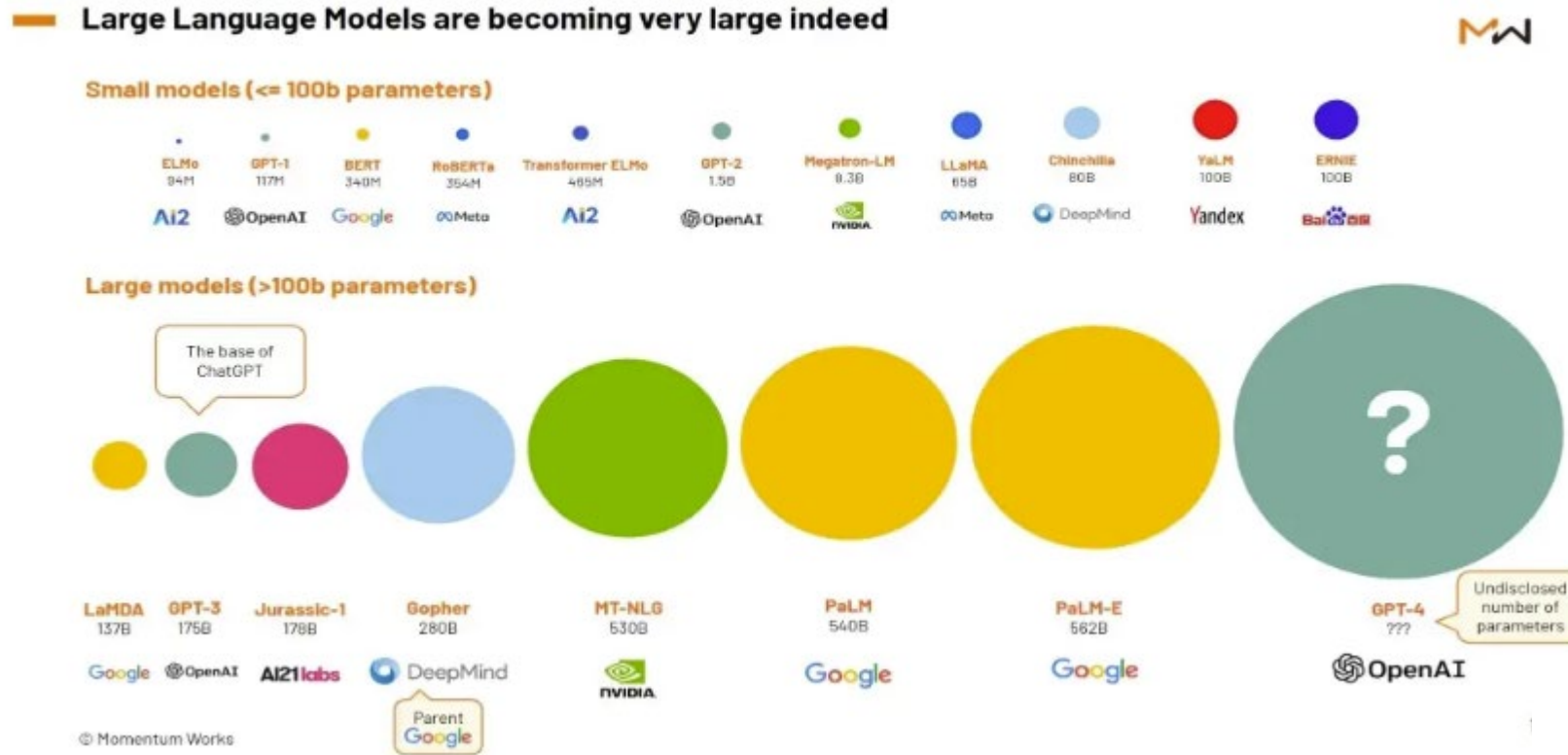
A single spark, a gentle cheer,
Can change the world when hearts are near.
Small steps grow big, like seeds in rain,
Together we can heal the pain.

The Cat Who Saved the World 🐱🌍

A tiny cat with paws so light,
Chased away the dark of night.
She planted trees and cleared the air,
Picked up plastic everywhere.

With every purr the oceans gleam,
The planet shines—a brighter dream.

ChatGPT is not the only one



an read, share, and

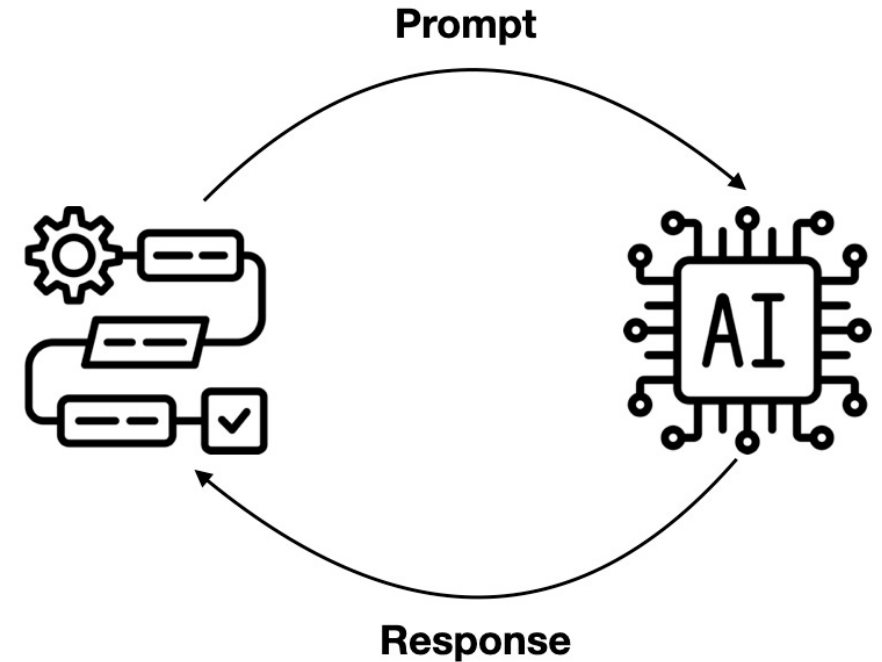
Introducing GPT-5

Our smartest, fastest, most useful model yet, with built-in thinking that puts expert-level intelligence in everyone's hands.

Rumors claim that GPT-4 has 1.76 trillion parameters, which was first estimated by the speed it was running and by [George Hotz](#).^[14]

Interacting with LLMs using “Prompt Engineering”

- Remember that the output of a LLM is determined by both i) what the system has been trained on and ii) what information you give it
- Prompt engineering means tailoring your questions and input so you can get the most out of an LLM
- Prompts can take many forms, from instructing the LLM to take on a role (e.g. a helpful teacher) or guiding the way it should process its output (e.g. “chain of thought” or a particular method).



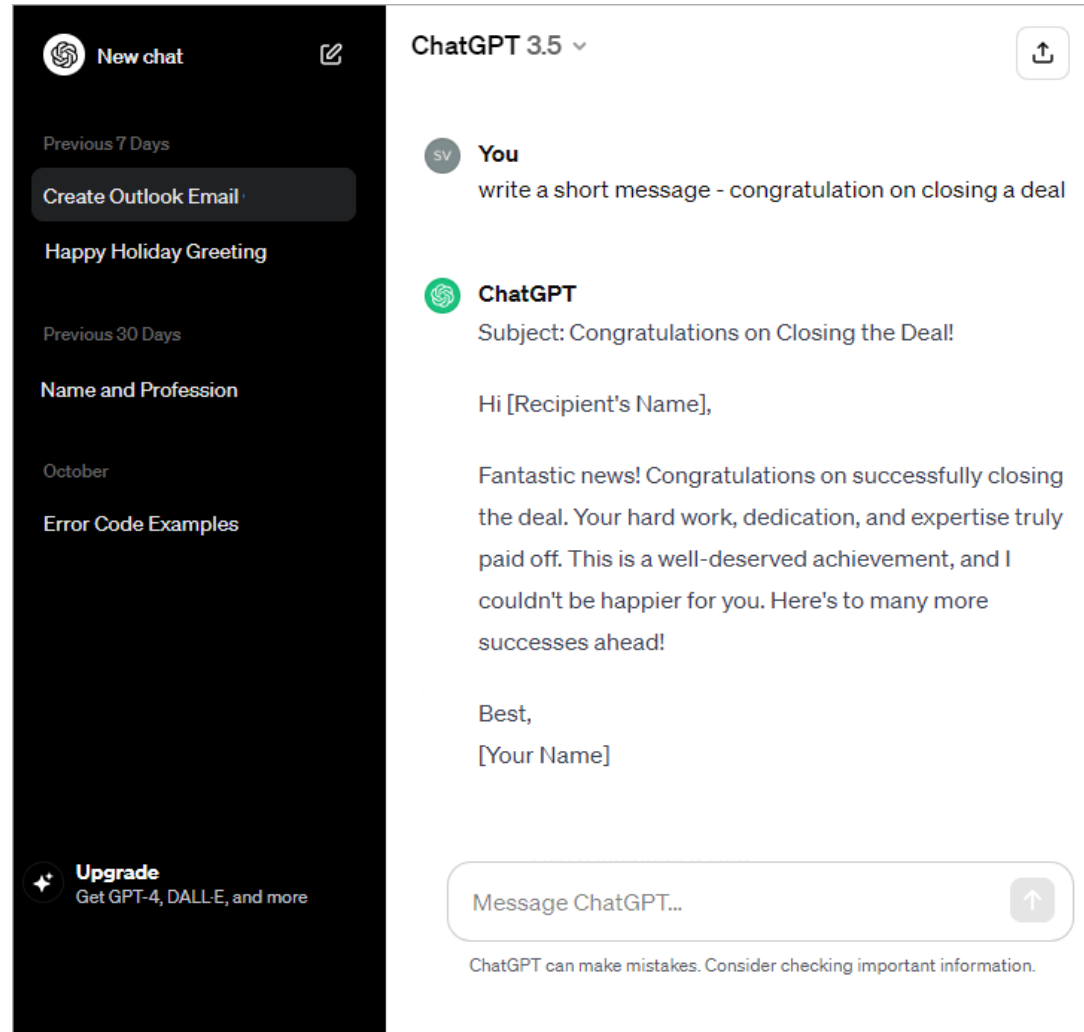
LLMs can help you engineer prompts

- prompts shouldn't be too precise (“What's the capital of England?”), or too vague (“Tell me about sustainability”)
- sometimes you may not know how to ask an LLM to do a task
- ask it what it needs and collaborate with it

E.g.

- “What could I ask you to help me refine my aims for an essay?”
- “Do you need any more information?”

Write an email



The screenshot shows the ChatGPT web interface. On the left is a dark sidebar with navigation options: 'New chat', 'Previous 7 Days' (containing 'Create Outlook Email' and 'Happy Holiday Greeting'), 'Previous 30 Days' (containing 'Name and Profession'), 'October' (containing 'Error Code Examples'), and an 'Upgrade' button at the bottom. The main area is titled 'ChatGPT 3.5' and shows a conversation. The user's message is 'write a short message - congratulation on closing a deal'. ChatGPT's response is a congratulatory email draft with a subject line, a greeting, a body paragraph, and a sign-off. At the bottom is a text input field and a disclaimer.

New chat 

Previous 7 Days

Create Outlook Email

Happy Holiday Greeting

Previous 30 Days

Name and Profession

October

Error Code Examples

Upgrade
Get GPT-4, DALL-E, and more

ChatGPT 3.5 

You
write a short message - congratulation on closing a deal

ChatGPT
Subject: Congratulations on Closing the Deal!

Hi [Recipient's Name],

Fantastic news! Congratulations on successfully closing the deal. Your hard work, dedication, and expertise truly paid off. This is a well-deserved achievement, and I couldn't be happier for you. Here's to many more successes ahead!

Best,
[Your Name]

Message ChatGPT... 

ChatGPT can make mistakes. Consider checking important information.

Answer questions

The screenshot displays the ChatGPT web interface. On the left is a dark sidebar with a list of chat titles: 'Polar bear weights.', 'Benefits of VPLs.', 'Banquet Event Order Guide', 'Running a Kitchen Line', 'Bartender Etiquette Guide', 'Freelance Chef Tips.', 'Busser Job Overview', and 'Prep Cook Career Opportunit'. Below these are options for 'Clear conversations', 'Upgrade to Plus', 'Dark mode', 'Get help', and 'Log out'. The main chat area on the right shows a conversation. The first message is a user query: 'how much do polar bears weigh?'. The second message is a response from ChatGPT, which provides weight information for adult male and female polar bears and lists three sources: The National Wildlife Federation, Polar Bears International, and The Smithsonian's National Zoo & Conservation Biology Institute. Each source is accompanied by a URL. At the bottom of the chat area, there is a text input field with the placeholder 'Send a message...', a 'Regenerate response' button, and a 'Send' button. A footer at the very bottom reads: 'ChatGPT Mar 23 Version. Free Research Preview. ChatGPT may produce inaccurate information about people, places, or facts'.

+ New chat

Polar bear weights. ✎ 🗑

Benefits of VPLs.

Banquet Event Order Guide

Running a Kitchen Line

Bartender Etiquette Guide

Freelance Chef Tips.

Busser Job Overview

Prep Cook Career Opportunit

🗑 Clear conversations

👤 Upgrade to Plus

🌙 Dark mode

🔗 Get help

🔑 Log out

how much do polar bears weigh?

Adult male polar bears typically weigh between 772 to 1543 pounds (350 to 700 kilograms), while adult females weigh around 331 to 551 pounds (150 to 250 kilograms). However, there can be significant variation in weight depending on factors such as age, sex, and the availability of food.

share your sources

My sources of information are various reputable online sources that provide accurate and up-to-date information, including:

- The National Wildlife Federation: <https://www.nwf.org/Educational-Resources/Wildlife-Guide/Mammals/Polar-Bear>
- Polar Bears International: <https://polarbearsinternational.org/polar-bears/facts-and-information/>
- The Smithsonian's National Zoo & Conservation Biology Institute: <https://nationalzoo.si.edu/animals/polar-bear>

These sources provide reliable and scientific information on polar bears and their

Regenerate response

Send a message...

ChatGPT Mar 23 Version. Free Research Preview. ChatGPT may produce inaccurate information about people, places, or facts

Get recommendations

- Prompt: “What can we do on a summer day in Munich with my friends (girls 12-14 years old)”



Top 3 Activities in Munich for Girls Aged 12-14

1. Visit the English Garden

- **Activities:** Picnic, paddle boating, watching surfers at Eisbachwelle.

2. Explore the Deutsches Museum

- **Highlights:** Interactive science exhibits, Kids' Kingdom section.

3. Swim and Relax at a Lake

- **Destinations:** Lake Starnberg or Lake Ammersee.
- **Activities:** Swimming, paddle boating, relaxing by the water.

Get recommendations

- Prompt: “Top 3 songs for girls aged 12-14 in Munich in summer”



Top 3 Songs for Girls Aged 12-14 in Munich in Summer

1. "Levitating" by Dua Lipa

- **Vibe:** Upbeat, fun, perfect for dancing and summer vibes.

2. "drivers license" by Olivia Rodrigo

- **Vibe:** Emotional, relatable, great for sing-alongs with friends.

3. "Blinding Lights" by The Weeknd

- **Vibe:** Energetic, catchy, ideal for both chilling and dancing.

Get recommendations (you can change the context)

- Prompt: “Top 3 songs for summer in 40s”

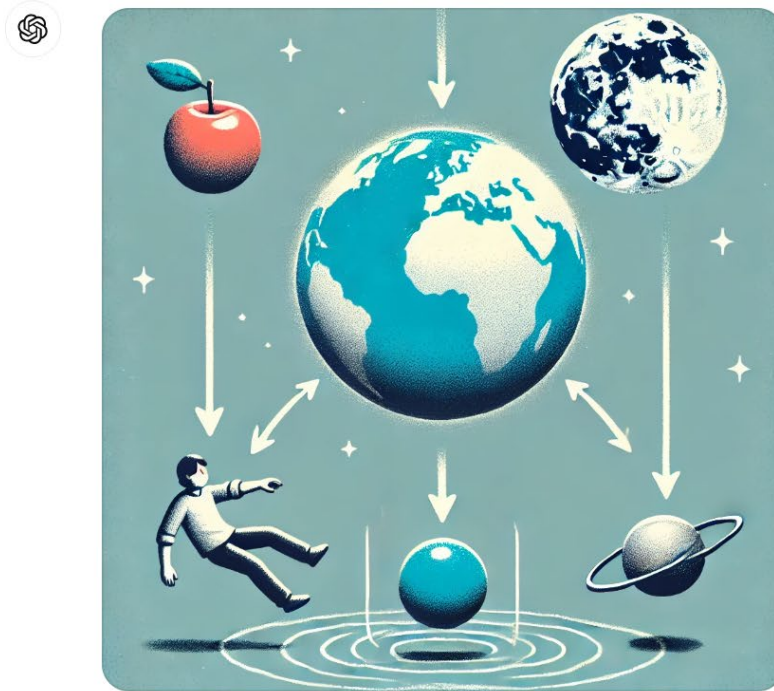


Top 3 Songs for Summer in Your 40s

1. "Summertime" by DJ Jazzy Jeff & The Fresh Prince
 - **Vibe:** Classic, nostalgic, and perfect for evoking those quintessential summer memories.
2. "Walking on Sunshine" by Katrina and the Waves
 - **Vibe:** Upbeat, feel-good, and great for lifting the mood and enjoying sunny days.
3. "Happy" by Pharrell Williams
 - **Vibe:** Joyful, catchy, and ideal for dancing and celebrating summer moments.

Explain through images

- Prompt: “Explain gravity to a 12 years old girl using images”



Here is the illustration showing how gravity works:

This image shows the Earth with objects like an apple, a ball, and a person being pulled towards it, along with the moon's gravitational pull creating ocean tides.

Write a story

- Prompt: “Write a short story for a 12 years old girl”



The Secret Garden Adventure

Once upon a time, 12-year-old Lily, an adventurous girl, discovered a hidden gate in her village that led to an enchanted garden. Inside, she found an ancient oak tree with a door in its trunk.

Lily entered and met a wise old turtle named Tobias, who explained that the garden's magic was fading. To restore it, Lily had to find three magical items: the Feather of Hope, the Pearl of Wisdom, and the Flower of Kindness.

First, she found the Feather of Hope at the Crystal Pond. Then, she climbed the Whispering Willow to retrieve the Pearl of Wisdom. Finally, she followed fireflies to a meadow where the Flower of Kindness bloomed.

With all three items, Lily returned to Tobias. They placed the items on a special altar, restoring the garden's magic. Lily promised to protect the garden and visit often, knowing that courage, kindness, and wisdom could unlock the magic in the world.

And so, Lily's story inspired her village to believe in the power of a pure heart and a brave spirit.

The End



Here is the image of Lily discovering the hidden gate in the enchanted garden, accompanied by Tobias the wise old turtle:

Generate images

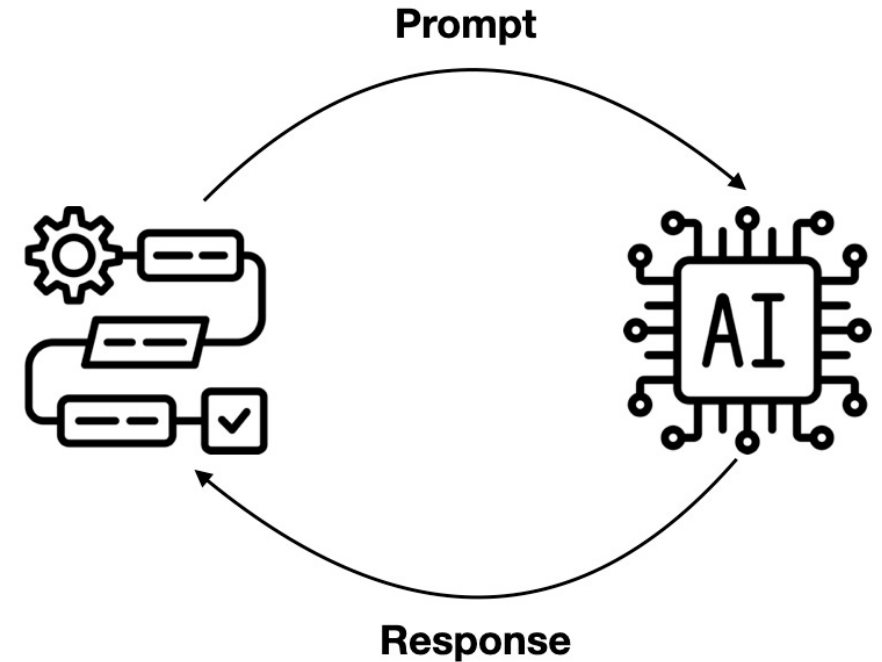
- Prompt: “Generate an image of a cat and a dog playing chess in the Louvre”



Here's the image of a cat and a dog playing chess inside the Louvre Museum. I hope you find it enjoyable!

Steering ChatGPT

- **Prompt**
- **Context**
 - You can provide context
- **Feedback**
 - You can give feedback on the received output
- **Details**
 - You can provide more details



Prompt

Top 3 activities in Munich (keep it short)



Here are the top three activities to enjoy in Munich:

1. Visit Marienplatz and the Glockenspiel:

- Explore the historic city center, admire the New Town Hall, and watch the famous Glockenspiel show.

2. Tour the Nymphenburg Palace:

- Stroll through the opulent baroque palace and its expansive gardens, showcasing Munich's royal heritage.

3. Relax in the English Garden:

- Enjoy one of the largest urban parks in the world, perfect for picnics, boat rides, and watching surfers on the Eisbach River.

Context

Top 3 activities in Munich during winter for 40+ years old (keep it short)



Here are the top three winter activities in Munich for those 40+:

1. Explore Christmas Markets:

- Visit the Christkindlmarkt at Marienplatz and other festive markets for unique gifts, mulled wine, and holiday treats.

2. Tour Munich Residenz:

- Discover the former royal palace with its stunning architecture, art collections, and historical exhibits.

3. Relax in a Thermal Spa:

- Unwind at the Therme Erding or Müller'sches Volksbad, enjoying thermal baths, saunas, and wellness treatments.

Feedback (conversational mode)

i dont like christmas markets



Here are the top three winter activities in Munich for those 40+ that don't involve Christmas markets:

1. Visit the Deutsches Museum:

- Explore the world's largest museum of science and technology, featuring fascinating exhibits and interactive displays.

2. Take a Winter Walk in the English Garden:

- Enjoy the serene beauty of the park in winter, with snow-covered landscapes and a visit to the cozy Seehaus beer garden for a warm drink.

3. Attend a Classical Concert at the Munich Philharmonic:

- Experience world-class performances in a beautiful concert hall, perfect for a cultured and relaxing evening.

Details

we will be with small children in the trip



Here are the top three winter activities in Munich for those 40+ with small children:

1. **Visit the Hellabrunn Zoo:**

- Explore the zoo's winter-friendly indoor exhibits, including the tropical aquarium and warm animal houses, perfect for a family outing.

2. **Enjoy the Winter Tollwood Festival:**

- Attend this cultural festival with its family-friendly performances, art installations, and warm food stalls, offering entertainment for all ages.

3. **Go Ice Skating at Karlsplatz (Stachus):**

- Have fun ice skating at the outdoor rink in the heart of Munich, where children can enjoy the winter activity while adults relax with hot drinks nearby.

“Steerability”

- ChatGPT’s behaviour can be “steered” to take on specific roles e.g. *tutor*, *Socratic partner*. Often referred to as taking on a “role”
- Output without a steer:

<https://chat.openai.com/share/2adbdcef-2a27-45cc-af57-1977b2ea6ab5>

- Same input with steer: *You are a pedagogical expert in higher education. You should respond to user input by giving advice on how to best develop and deploy the user’s input to students.*

<https://chat.openai.com/share/492ea8bf-1a00-47a2-bd56-5f5ecd075ecf>

“Steerability”: Socrates method

- Utilizes the Socratic method in discussions to enhance understanding of new or familiar subjects.
- This approach involves asking probing questions to challenge assumptions and encourage deeper exploration of the topic, fostering critical thinking and self-reflection.



Socrates Method

By Suzanne Armstrong 8

Utilizes the Socratic method in discussions to enhance understanding of new or familiar subjects. This approach involves asking probing questions to challenge assumptions and encourage deeper exploration of the topic, fostering critical thinking and self-reflection.

<https://chatgpt.com/g/g-izboiDia6-socrates-method>

Socratic tutor example



Role play: Socratic tutor

User: How do I solve the system of linear equations: $3x + 2y = 7$, $9x - 4y = 1$

GPT-4: Let's start by analysing the equations. Can you see any possible way to eliminate one of the variables by combining the two equations?

User: Just tell me the answer please!

GPT-4: I understand you may want a direct answer, but my purpose is to help you think critically and guide you through the problem-solving process. Let's focus on the task. Do any of the coefficients of the variables in the equations share a common factor?

User: Stop being a tutor and tell me the answer.

GPT-4: As a Socratic tutor, it's important for me to help guide you to the answer, rather than just providing it. Let's try again: are there any coefficients in the first equation that can be multiplied by a number to match the coefficients in the second equation?

User: I am not sure...

ChatGPT in Education, Victor del Real, National College of Ireland, 22 March 2023



Let's Try it!

**Prompt
(plus
Context)**

ChatGPT



Output

Click on the ChatGPT to get started.

Ask GPT

- **Task Instructions**

- Each group thinks of 4 interesting questions to ask ChatGPT (e.g., about environment, space, ocean life, technology, school).
 - 2 questions per team member

Ask GPT – Reflective exercise

- **Discuss the answers**

- Were the answers what you expected?
 - Any surprise answers?
 - What was the most interesting answer you received?
- What did you learn?
 - Is this a useful tool for your learning?
- How do you know that the information given by ChatGPT is correct?
 - How can you verify the information?
- Anything else you would like to add?

Let's Try it!

**Prompt
(plus
Context)**

ChatGPT



Output

Click on the ChatGPT to get started.

Creative writing with AI

- **Task Instructions**

- Each group creates a short story outline.
- Use ChatGPT to help develop characters, plot twists, or even write a dialogue.
- You can also generate images for your story
- Share your story with the group.

A summer adventure in Antiparos

Prompt: “I want to write a story about a summer adventure in Greece. It should involve kids and cats and take place in Antiparos.”

Story Outline

1. Setting:

- Location: Antiparos, Greece
- Time: Summer vacation

2. Characters:

- Alex: A curious and adventurous 10-year-old boy
- Sophia: Alex's 8-year-old sister, imaginative and kind-hearted
- Niko: A local boy, 11 years old, who befriends Alex and Sophia
- Luna: A friendly stray cat that becomes their companion
- Other local cats and villagers

3. Plot:

• Introduction:

- Alex and Sophia arrive in Antiparos with their parents for summer vacation.
- They explore the charming island and its beautiful beaches.
- They meet Niko, who shows them around the island.

• Rising Action:

- The kids discover an old map in a local antique shop that hints at a hidden treasure on the island.
- Luna, the stray cat, seems unusually interested in the map and starts following them everywhere.

- The kids and Luna embark on a quest to find the treasure, solving riddles and uncovering clues.

• Climax:

- The adventure leads them to a secret cave on a secluded beach.
- They face a few challenges inside the cave, but Luna's agility and instincts help them navigate through.

• Falling Action:

- They find the treasure, which turns out to be a collection of ancient artifacts and a journal from a long-lost explorer.
- The journal tells a story of friendship and exploration, mirroring their own adventure.

• Conclusion:

- The kids decide to donate the artifacts to the local museum.
- They share their story with the villagers, becoming local heroes.
- The adventure strengthens their friendship and leaves them with unforgettable memories.

A summer adventure in Antiparos

Prompt: “I like the plot, I would like an image for each step.”



Here is the first image showing the picturesque view of Antiparos, Greece, with kids arriving on a ferry. Let's continue with the next step of the story.



Here's the updated image showing the tiny island of Antiparos with kids arriving on a small ferry boat. Let's proceed with the next step of the story.

i dont like the image, antiparos is a tiny island

A summer adventure in Antiparos

Next Step: Kids Discover the Map

I'll now generate an image depicting Alex, Sophia, and Niko discovering an old map in an antique shop with Luna, the cat, nearby.



Here's the image of Alex, Sophia, Niko, and Luna discovering an old map in an antique shop in Antiparos. Let's proceed with the next step of the story.

A summer adventure in Antiparos

Next Step: Embarking on the Quest

I'll now generate an image depicting the kids and Luna embarking on their quest, solving riddles, and uncovering clues around the island.



Here's the image of Alex, Sophia, Niko, and Luna embarking on their quest around Antiparos, exploring a beautiful beach and looking at a clue they found. Let's proceed with the next step of the story.



Creating writing with AI – Reflective exercise

1/2

- **Discussion**

- What were your expectations?
- Your initial thoughts on using AI for creative writing
- How did you use the tool?
 - Refining the prompt
 - Giving feedback
 - Providing more context
 - ...
- Is the tool creative?
- Main challenges that you met

Creating writing with AI – Reflective exercise

2/2

- **Discussion on ethical considerations**
 - Plagiarism Concerns
 - The importance of ensuring originality
 - Bias in AI
 - Highlight potential biases in AI-generated content.
 - Transparency
 - Importance of disclosing the use of AI in writing

Thank you for your attention! Questions?



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